MTH101, Basic Linear Algebra

Monsoon 2025

11/09/2025, 2:00PM-3:00PM.

Tutorial-2

(1) Find all possible line equations passing through $(0, 0) \in \mathbb{R}^2$, where $\mathbb{R}$ is the set of all real numbers.

(2) Write the matrix equation of the system

$$
\begin{aligned}
3X_1 + 2X_2 + X_3 &= 5 \\
X_1 - 4X_3 &= 4 \\
2X_1 - 5X_2 + X_3 &= 0
\end{aligned}
$$

(3) Let $I_n := (a_{ij})$ be the $n \times n$ matrix, called the *identity matrix*, such that $a_{ii} = 1$ and $a_{ij} = 0$ for $i \neq j$. Solve the matrix system

$$
I_n X = B, \quad \text{where } X = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{pmatrix}, \quad B := \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}, \; b_1, \ldots, b_n \in \mathbb{R}.
$$

(4) Show that for

$$
A := \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \quad \text{and} \quad B := \begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix},
$$

$AB = BA = I_2$.

(5) Solve the system of equations

$$
\begin{aligned}
X + 2Y &= 2 \\
3X + 4Y &= -4
\end{aligned}
$$

by converting it into a matrix equation and by using problems 3 and 4.